

```
-- CONTROL FLOW STATEMENTS IN STORED PROCEDURES:-  
-- 1) IF/ ELSE IN STORED PROCEDURES
```

```
-- SYNTAX
```

```
/*IF CONDITION THEN  
        STATEMENTS;  
ELSE IF CONDITION THEN  
        STATEMENTS;  
ELSE  
        STATEMENTSS;  
END IF;*/
```

```
DELIMITER $$
```

```
CREATE PROCEDURE SALARY_CATEGORY(IN SALARY DECIMAL(9,2))
```

```
BEGIN
```

```
IF SALARY > 90000 THEN
```

```
        SELECT "HIGH EARNER" AS SALARY_CATEGORY;
```

```
ELSEIF SALARY BETWEEN 60000 AND 89999 THEN
```

```
        SELECT "MEDIUM EARNER" AS SALARY_CATEGORY;
```

```
ELSE
```

```
        SELECT "LOW EARNER" AS SALARY_CATEGORY;
```

```
END IF;
```

```
END $$
```

```
DELIMITER ;
```

```
DROP PROCEDURE SALARY_CATEGORY;
```

```
CALL SALARY_CATEGORY(34000);
```

```
-- 2) CASE STATEMENTS:-
```

```
-- CALL CHECK_DAY(1);    -- > MONDAY
```

```
-- SWITCH CASE
```

```
DELIMITER //
```

```
CREATE PROCEDURE CHECK_DAY(IN DAY_NUMBER INT)
```

```
BEGIN
```

```
        CASE DAY_NUMBER
```

```
                WHEN 1 THEN SELECT 'MONDAY';
```

```
                WHEN 2 THEN SELECT 'TUESDAY';
```

```
                WHEN 3 THEN SELECT 'WEDNESDAY';
```

```
                WHEN 4 THEN SELECT 'THURSDAY';
```

```
                WHEN 5 THEN SELECT 'FRIDAY';
```

```
                WHEN 6 THEN SELECT 'SATURDAY';
```

```
                WHEN 7 THEN SELECT 'SUNDAY';
```

```
                ELSE SELECT 'INVALID DAY';
```

```
END CASE;
```

```
END //
```

```
DELIMITER ;
```

```
CALL CHECK_DAY(234);
```

```

-- CONDITONAL CASE
DELIMITER $$
CREATE PROCEDURE SALARY_CATEGORY_CASE(IN SALARY DECIMAL(9,2))
BEGIN
CASE
WHEN SALARY > 90000 THEN
    SELECT "HIGH EARNER" AS SALARY_CATEGORY;
WHEN SALARY BETWEEN 60000 AND 89999 THEN
    SELECT "MEDIUM EARNER" AS SALARY_CATEGORY;
ELSE
    SELECT "LOW EARNER" AS SALARY_CATEGORY;
END CASE;
END $$
DELIMITER ;

CALL SALARY_CATEGORY_CASE(100000);

```

```

-- LOOPS:-
-- WHILE LOOP:-
-- SYNTAX :-
/*
WHILE CONDITION DO
    STATMENTS;
END WHILE;
*/

```

```

-- PRINT NUMBER FROM 1 TO 5
DELIMITER $$
CREATE PROCEDURE PRINT_NUMBERS()
BEGIN
    declare i int default 1; -- declartion of variable
    while i <= 5 do -- condition
        select i;      -- statments
        set i = i + 1; -- increment
    end while;
end $$
delimiter ;

```

```

    call print_numbers();

```

```

-- loop(generic loop):-

```

```

-- syntax:-
/*
loopName : loop
    statments;
if condition then

```

```

        leave loopName;
end if;
end loop;
*/

delimiter $$
create procedure loop_example()
begin
    DECLARE i int default 1;

    myLoop : loop
    select i;
    set i = i + 1;
    if i >= 10 then
    leave myLoop;
    end if;
    end loop;
end $$
delimiter ;

call loop_example();

-- repeat loop:-
-- runs atleast once
-- syntax :-
/*
repeat
statements;
until condition
end repeat;
*/
delimiter $$
create procedure repeat_example()
begin
declare i int default 1;
repeat
select i;
set i = i + 1;
until i > 1
end repeat;
end $$
delimiter ;
call repeat_example();

drop procedure repeat_example;

select * from employees;

delimiter $$

```

```

create procedure find_high_salary(in target_salary decimal(9,2))
begin
    declare i int default 1; -- emp_id
    declare emp_salary decimal(9,2); -- emp_salary
    myLoop : loop -- generic loop

    select salary into emp_salary -- 40000
    from employees
    where emp_id = i; -- statment

    if emp_salary > target_salary then -- condition 70000 > 60000
    select 'High salary employee found';
    leave myLoop;
    end if;
    set i = i + 1; -- increment
    end loop;
end $$
delimiter ;

call find_high_salary(90000);

```

```

select * from employees;

```

```

drop procedure find_high_salary;

```

```

delimiter $$
create procedure find_high_salary(in target_salary decimal(9,2))
begin
    declare i int default 1; -- id 2
    declare emp_salary decimal; -- 55000
    declare found_employee boolean default false; -- false
    declare total int; -- 11

    select count(*) into total from employees;

myLoop : loop
    if i > total then -- exit conditon
    leave myLoop;
    end if;

    select salary into emp_salary
    from employees
    where emp_id = i;

    if emp_salary > target_salary then

```

```

select 'High salary employees found' as 'Salary status';
set found_employee = true;
leave myLoop;
end if;
set i = i + 1;
end loop;

```

```

if found_employee = false then
select 'High salaried employee not found';
end if;
end $$
delimiter ;

```

```

call find_high_salary(60000);

```

```

select * from employees;

```

```

DELIMITER $$
CREATE PROCEDURE FOUND_HIGH_SALARY_EMPLOYEE(IN TARGET_SALARY DECIMAL(9,2))
BEGIN
    -- DECLARING THE VARIABLES
    DECLARE E_ID INT DEFAULT 1;
    DECLARE TOTAL_EMPLOYEES INT; -- 11
    DECLARE EMP_SALARY DECIMAL(9,2);
    DECLARE FOUND_EMPLOYEES BOOLEAN DEFAULT FALSE;

    SELECT COUNT(*) INTO TOTAL_EMPLOYEES
    FROM EMPLOYEES;

    myLoop : LOOP -- MYLOOP IS YOUR LOOP LABEL

    IF EMP_ID > TOTAL_EMPLOYEES THEN -- EXIT CONDITION
    LEAVE myLoop;
    END IF;

    SELECT SALARY INTO EMP_SALARY
    FROM EMPLOYEES
    WHERE EMP_ID = E_ID;

    IF EMP_SALARY > TARGET_SALARY THEN
    SELECT "HIGH SALARY EMPLOYEE FOUND";
    SET FOUND_EMPLOYEES = TRUE;
    LEAVE myLoop;
    END IF;

    SET E_ID = E_ID + 1;
    END LOOP;

```

```
IF FOUND_EMPLOYEES = FALSE THEN
SELECT "NO HIGH SALARIED EMPLOYEE FOUND!";
END IF;
```

```
END $$
DELIMITER ;
```

```
-- WITH WHILE LOOP
```

```
DELIMITER $$
CREATE PROCEDURE FOUND_HIGH_SALARY_EMPLOYEE(IN TARGET_SALARY DECIMAL(9,2))
BEGIN
    -- DECLARING THE VARIABLES
    DECLARE E_ID INT DEFAULT 1;
    DECLARE TOTAL_EMPLOYEES INT; -- 11
    DECLARE EMP_SALARY DECIMAL(9,2);
    DECLARE FOUND_EMPLOYEES BOOLEAN DEFAULT FALSE;

    SELECT COUNT(*) INTO TOTAL_EMPLOYEES
    FROM EMPLOYEES;

    WHILE_LOOP : WHILE E_ID <= TOTAL_EMPLOYEES DO -- WHILE_LOOP IS YOUR LOOP LABEL

        SELECT SALARY INTO EMP_SALARY
        FROM EMPLOYEES
        WHERE EMP_ID = E_ID;

        IF EMP_SALARY > TARGET_SALARY THEN
            SELECT "HIGH SALARY EMPLOYEE FOUND";
            SET FOUND_EMPLOYEES = TRUE;
            LEAVE WHILE_LOOP;
        END IF;

        SET E_ID = E_ID + 1;
    END WHILE;

    IF FOUND_EMPLOYEES = FALSE THEN
        SELECT "NO HIGH SALARIED EMPLOYEE FOUND!";
    END IF;

END $$
DELIMITER ;
```

```
CALL FOUND_HIGH_SALARY_EMPLOYEE(40000);
```

```

--
-- UPDATE EMPLOYEES
-- GIVE A HIKE OF 20 PERCENT FOR EMPLOYEE EARNING LESS THAN 40000
-- GIVE A HIKE OF 10 PERCENT FOR EMPLOYEE EARNING MORE THAN 40000
-- AND LESS THAN 8000
-- GIVE A HIKE OF 5 PERCENT FOR EMPLOYEE EARNING MORE THAN 80000

DELIMITER $$
CREATE PROCEDURE HIKE_SALARY()
BEGIN
    -- VARIABLES DECLARE
    DECLARE E_ID INT DEFAULT 1;
    DECLARE E_SALARY DECIMAL(9,2);
    DECLARE TOTAL_EMPLOYEES INT;
    DECLARE NEW_SALARY DECIMAL(9,2);

    SELECT COUNT(*) INTO TOTAL_EMPLOYEES
    FROM EMPLOYEES;

    WHILE_LOOP : WHILE E_ID <= TOTAL_EMPLOYEES DO

        SELECT SALARY INTO E_SALARY
        FROM EMPLOYEES
        WHERE EMP_ID = E_ID;

        IF E_SALARY >= 80000 THEN
            SET NEW_SALARY = E_SALARY + (E_SALARY * 0.05);
        ELSEIF E_SALARY BETWEEN 40000 AND 79999 THEN
            SET NEW_SALARY = E_SALARY + (E_SALARY * 0.1);
        ELSE
            SET NEW_SALARY = E_SALARY + (E_SALARY * 0.2);
        END IF;

        UPDATE EMPLOYEES
        SET SALARY = NEW_SALARY
        WHERE EMP_ID = E_ID;

        SET E_ID = E_ID + 1;
    END WHILE;
END $$
DELIMITER ;

SELECT * FROM EMPLOYEES;
CALL HIKE_SALARY();

```